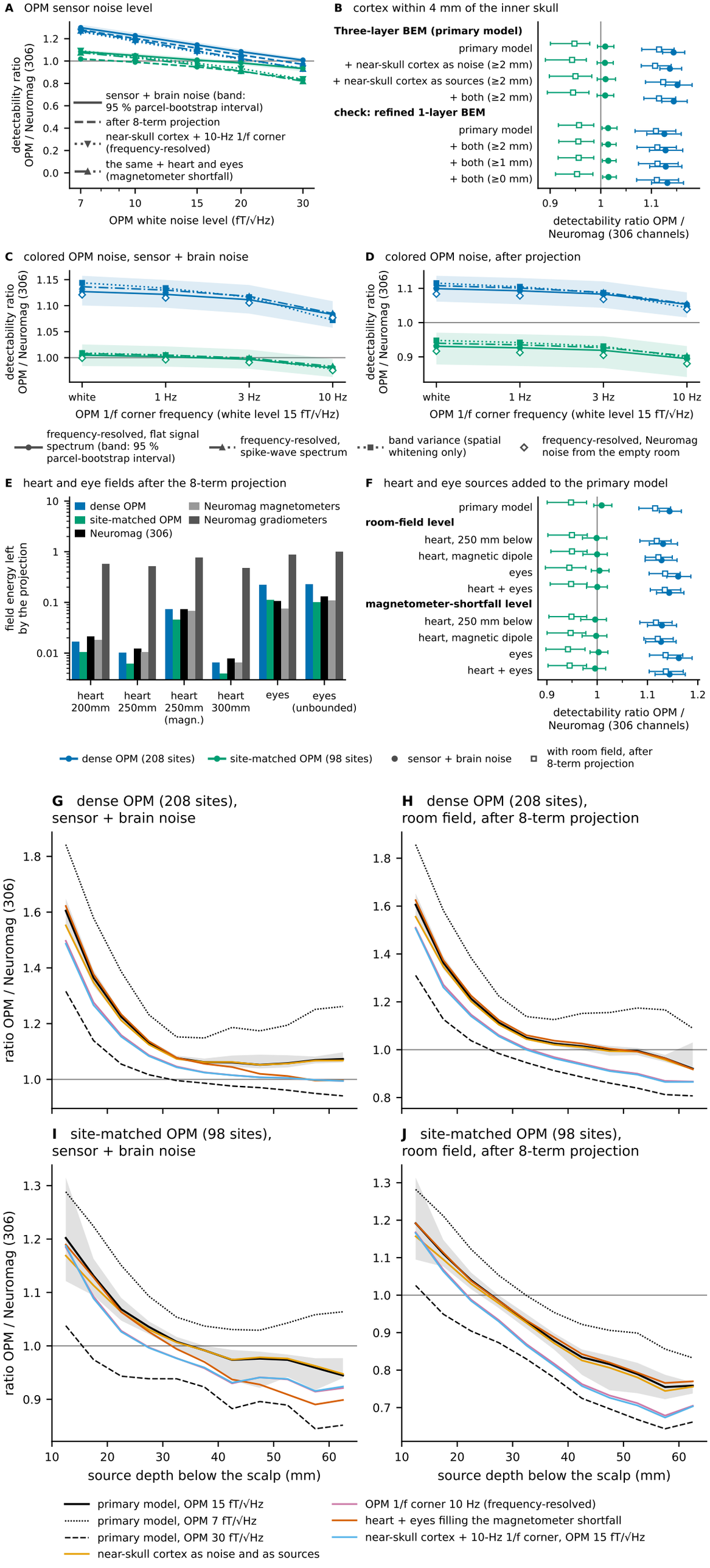

A OPM sensor noise level
detectability ratio OPM / Neuromag (306)
sensor + brain noise (band: 95 % parcel-bootstrap interval)
after 8-term projection
near-skull cortex + 10-Hz 1/f corner (frequency-resolved)
the same + heart and eyes (magnetometer shortfall)
OPM white noise level (fT/√Hz)
B cortex within 4 mm of the inner skull
Three-layer BEM (primary model)
primary model
+ near-skull cortex as noise (≥2 mm)
+ near-skull cortex as sources (≥2 mm)
+ both (≥2 mm)
check: refined 1-layer BEM
primary model
+ both (≥2 mm)
+ both (≥1 mm)
+ both (≥0 mm)
detectability ratio OPM / Neuromag (306 channels)
C colored OPM noise, sensor + brain noise
D colored OPM noise, after projection
white
1 Hz
3 Hz
10 Hz
OPM 1/f corner frequency (white level 15 fT/√Hz)
frequency-resolved, flat signal spectrum (band: 95 % parcel-bootstrap interval)
frequency-resolved, spike-wave spectrum
band variance (spatial whitening only)
frequency-resolved, Neuromag noise from the empty room
E heart and eye fields after the 8-term projection
dense OPM
site-matched OPM
Neuromag (306)
Neuromag magnetometers
Neuromag gradiometers
field energy left by the projection
heart 200mm
heart 250mm
heart 250mm (magn.)
heart 300mm
eyes
eyes (unbounded)
F heart and eye sources added to the primary model
primary model
room-field level
heart, 250 mm below
heart, magnetic dipole
eyes
heart + eyes
magnetometer-shortfall level
heart, 250 mm below
heart, magnetic dipole
eyes
heart + eyes
detectability ratio OPM / Neuromag (306 channels)
dense OPM (208 sites)
site-matched OPM (98 sites)
sensor + brain noise
with room field, after 8-term projection
G dense OPM (208 sites), sensor + brain noise
H dense OPM (208 sites), room field, after 8-term projection
I site-matched OPM (98 sites), sensor + brain noise
J site-matched OPM (98 sites), room field, after 8-term projection
ratio OPM / Neuromag (306)
source depth below the scalp (mm)
primary model, OPM 15 fT/√Hz
primary model, OPM 7 fT/√Hz
primary model, OPM 30 fT/√Hz
near-skull cortex as noise and as sources
OPM 1/f corner 10 Hz (frequency-resolved)
heart + eyes filling the magnetometer shortfall
near-skull cortex + 10-Hz 1/f corner, OPM 15 fT/√Hz